CAR PRICE PREDICTION WITH MACHINE LEARNING

CodeAlpha Data Science Internship | Task 3

Project Report

Project	Car Price Prediction
Dataset	car data.csv
Records before cleaning	301
Records after cleaning	299
Duplicate records removed	2
Target variable	Selling_Price

Prepared as part of the CodeAlpha Data Science Internship.

1. Project Overview

This project develops a machine-learning workflow to estimate the selling price of a used car from vehicle characteristics. The supplied dataset contains vehicle name, manufacturing year, selling price, present price, kilometres driven, fuel type, seller type, transmission type, and previous-owner count.

The workflow includes dataset inspection, cleaning, exploratory analysis, feature engineering, categorical encoding, model training, and evaluation. Linear Regression and Random Forest Regression are compared on a held-out test set.

2. Project Objectives

• Understand the structure and quality of the supplied car dataset.

• Clean duplicate records and prepare features for modeling.

• Explore how vehicle and seller characteristics relate to selling price.

• Build regression models to predict selling price.

• Compare models using MAE, RMSE, and R^2 metrics.

• Identify which input features contribute most to predictive performance.

3. Dataset Description

The original file contains 301 rows and 9 columns. The target variable is Selling_Price. The input attributes include Car_Name, Year, Present_Price, Driven_kms, Fuel_Type, Selling_type, Transmission, and Owner.

The initial inspection found 0 missing values and 2 exact duplicate rows. After removing duplicate records, 299 rows remained. No missing values were present in the supplied file.

Column	Description
Car_Name	Name or model identifier of the car
Year	Manufacturing year
Selling_Price	Observed selling price; prediction target
Present_Price	Present/listed price attribute in the dataset
Driven_kms	Distance driven in kilometres
Fuel_Type	Petrol, Diesel, or CNG
Selling_type	Dealer or Individual
Transmission	Manual or Automatic
Owner	Previous-owner category/count

4. Data Preprocessing and Feature Engineering

Column labels were standardized by replacing spaces with underscores, and whitespace around categorical values was trimmed. Exact duplicate records were removed. The original CSV was retained separately from the cleaned output.

A derived Vehicle_Age feature was created as the difference between the maximum manufacturing year in this dataset and each car's manufacturing year. This is a relative age feature within the supplied dataset, not an age calculated against the current calendar year.

Numerical columns were passed through a median imputer and categorical columns through a most-frequent imputer followed by one-hot encoding. The OneHotEncoder was configured to ignore unseen categories, allowing the pipeline to handle categories that may appear in future input rows.

5. Exploratory Data Analysis

The project generates plots for the distribution of selling prices, the relationship between present price and selling price, average selling price by fuel type, and actual versus predicted values. These charts provide a visual overview of the dataset and model performance.

Relationships in this dataset should be interpreted as associations, not proof that a feature independently causes a particular selling price. The dataset is relatively small and may not represent every used-car market or time period.

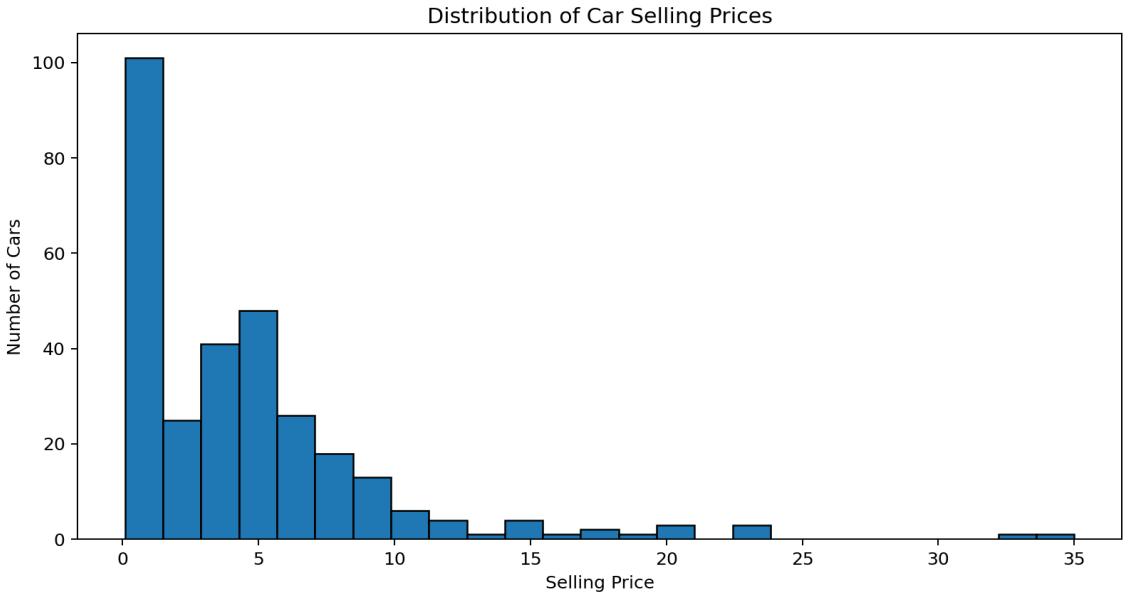


Figure 1. Distribution of observed selling prices.

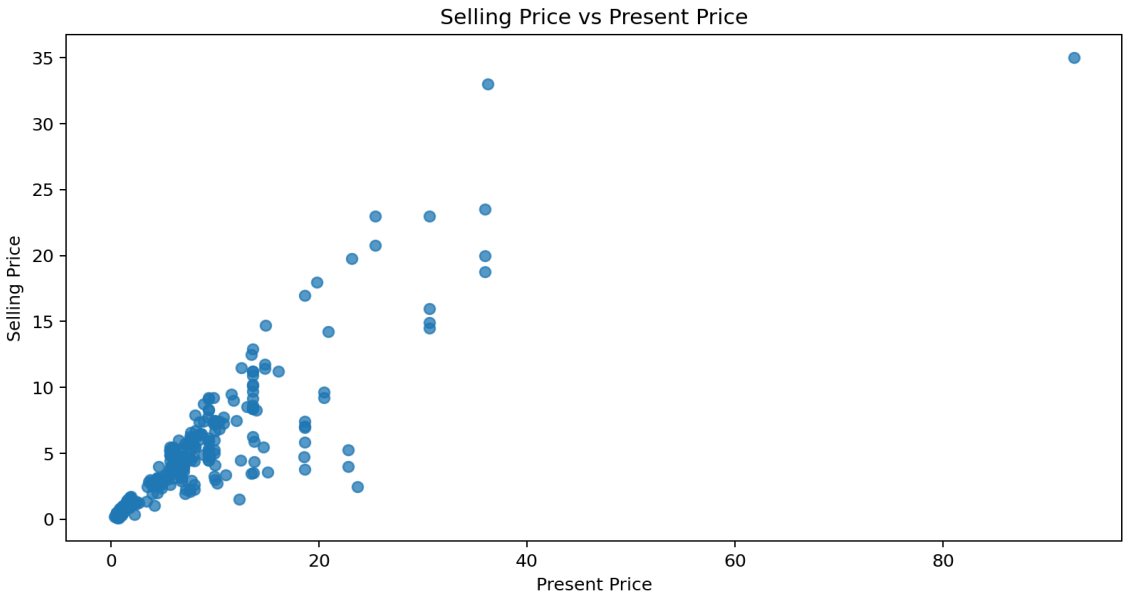


Figure 2. Relationship between present price and selling price.

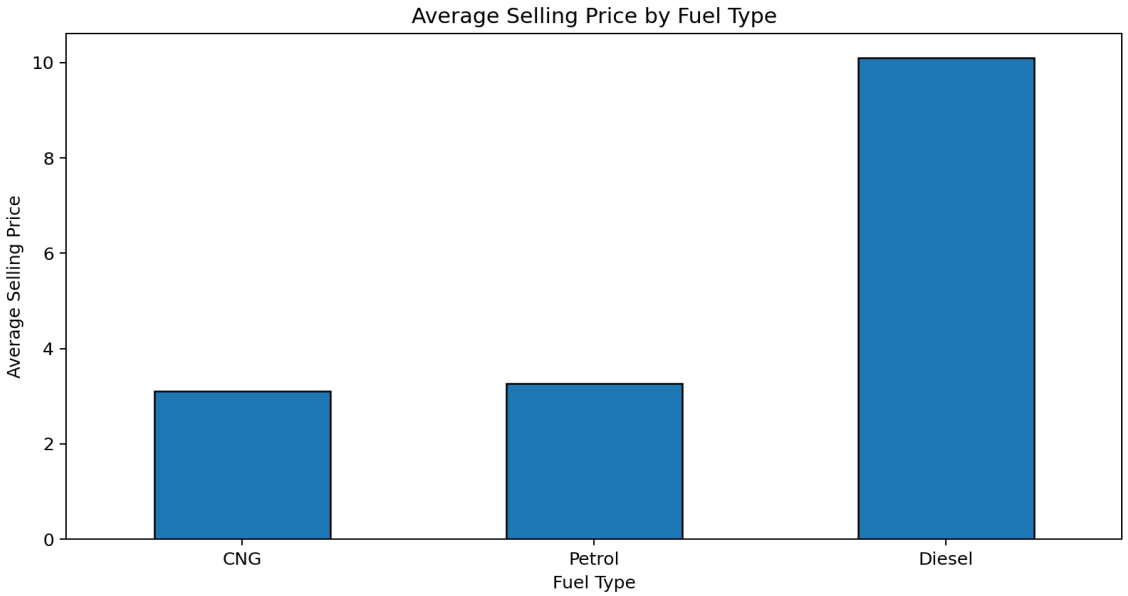


Figure 3. Average selling price by fuel type.

6. Machine Learning Methodology

The target is Selling_Price. The selected predictors are Car_Name, Vehicle_Age, Present_Price, Driven_kms, Fuel_Type, Selling_type, Transmission, and Owner. The data was divided into training and testing subsets using an 80:20 split with random_state=42.

Two regression algorithms were evaluated: Linear Regression as a baseline model and Random Forest Regressor as a non-linear ensemble model. Preprocessing and model fitting were combined in Scikit-learn Pipelines to ensure the transformations were learned from the training data only.

Model evaluation on the held-out test set

Model	MAE	RMSE	R^2
Linear Regression	1.4121	2.4316	0.7706
Random Forest Regressor	1.3688	3.2628	0.5869

7. Evaluation Metrics and Model Selection

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted prices. Root Mean Squared Error (RMSE) penalizes larger errors more strongly. R^2 indicates the proportion of target variance explained by the model on the test set; higher values generally indicate a better fit, while negative values indicate performance worse than predicting the test-set mean.

Based on the test-set R^2 score, Linear Regression was selected as the best-performing model in this experiment. Its test metrics were MAE 1.4121, RMSE 2.4316, and R^2 0.7706. This selection is specific to the given train/test split and should be validated further before deployment.

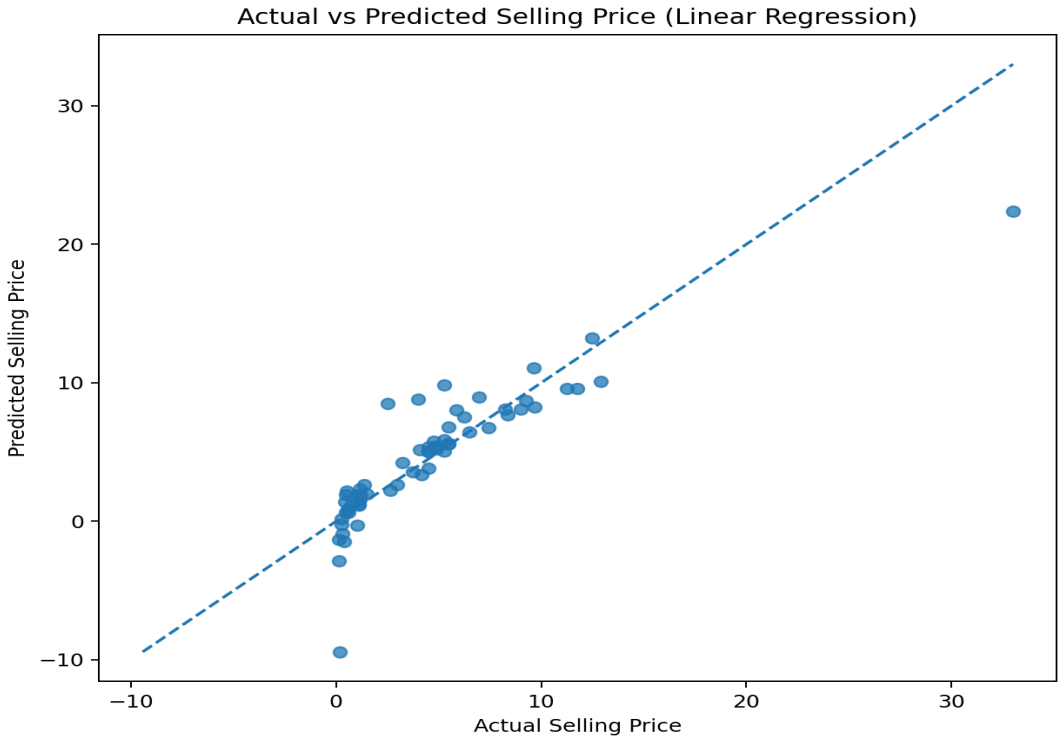


Figure 4. Actual versus predicted selling prices for the selected model.

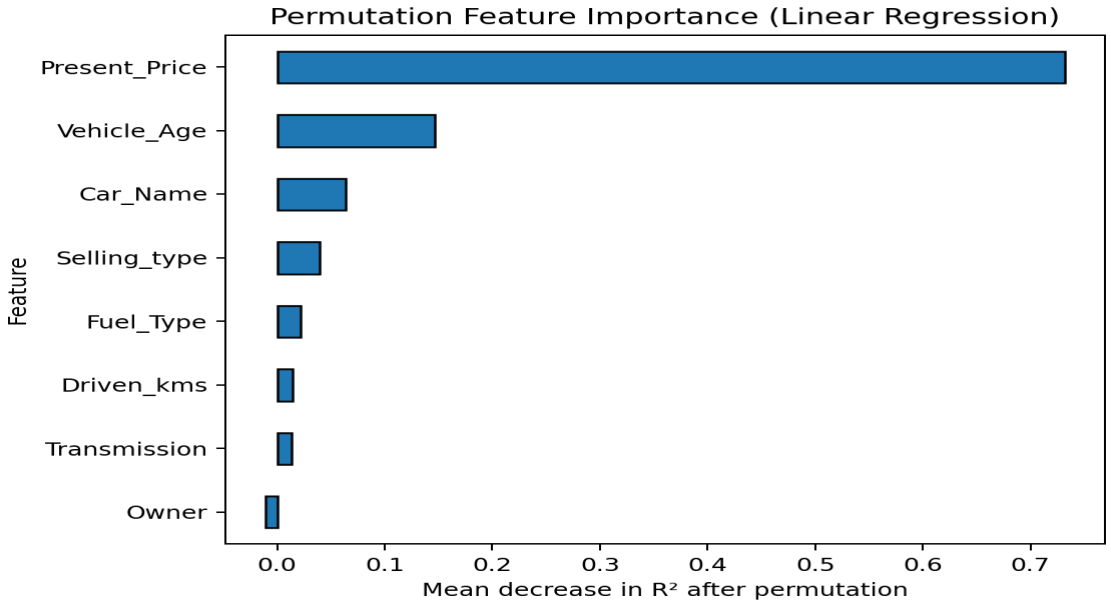


Figure 5. Permutation feature importance on the held-out test set.

8. Key Findings and Limitations

The cleaned dataset contains 299 records after removing 2 duplicate rows. The prediction workflow compares a linear baseline with a tree-based ensemble and reports error metrics on a separate test subset.

Permutation importance estimates how much the model's test R^2 changes when a feature is shuffled. It is a model-specific diagnostic and may be unstable with a small test sample or correlated features.

Important limitations include the small sample size, possible sampling bias, the absence of some potentially relevant attributes (for example, location, condition, service history, and exact variant), and the fact that prices are tied to the dataset's original market context. The model should not be treated as a live market valuation tool without updated data and further validation.

9. Conclusion

This project demonstrates an end-to-end used-car price prediction workflow using Python and Scikit-learn. It covers data cleaning, exploratory visualization, feature engineering, preprocessing pipelines, regression modeling, and model evaluation. The resulting notebook and scripts provide a reproducible starting point for further experimentation with larger and more representative vehicle datasets.

10. Tools and Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook, and ReportLab.